

Roll No.:

END SEMESTER EXAMINATION DECEMBER 2025

Name of the Program: B.Tech CSE

Semester: V

Name of the Course: Computer System Security

Course Code: TCS591

Time: 3 Hours

Maximum Marks: 100

Note:

- i. All questions are compulsory.
- ii. Answer any two sub questions among a, b & c in each main questions
- iii. Total marks in each main question are twenty.
- iv. Each questions carries 20 marks

Q1	(20Marks)	
(a)	Explain buffer overflow with an example. How does it lead to control hijacking?	CO1
(b)	Demonstrate how integer overflow can be exploited to bypass security checks in a simple C program.	CO1
(c)	Explain the working of static, dynamic, and concolic analysis tools. Provide an example scenario where each is useful.	CO1
Q2		
(a)	Demonstrate how a race condition vulnerability can be exploited using a symbolic link attack.	CO2
(b)	Explain the buffer overflow defense mechanisms (ASLR, DEP, stack canaries) and analyze their effectiveness.	CO2
(c)	Define Content Security Policy (CSP). How does it mitigate script injection attacks?	CO2
Q3		
(a)	Explain the role of browser sandboxing. How do modern browsers separate processes for security?	CO4
(b)	Explain the role of HTTPS in securing communications. Discuss certificate validation steps.	CO3
(c)	Consider a social media web app. Perform an attack-surface analysis identifying major security vulnerabilities.	CO3
Q4		
(a)	Describe the sandboxing model used in Android applications. How do permissions contribute to isolation?	CO5
(b)	Analyze differences in threat models for rooted vs non-rooted Android devices.	CO5
(c)	A smartphone begins sending unusual network traffic. Analyze possible spyware behavior and propose detection methods.	CO4
Q5		
(a)	Explain the difference between Meltdown and Spectre attacks based on speculative execution.	CO5
(b)	Describe authentication factors. Construct a secure password policy for a corporate environment.	CO6
(c)	Define Access Control Lists (ACLs). How do they differ from role-based access control (RBAC)?	CO6

